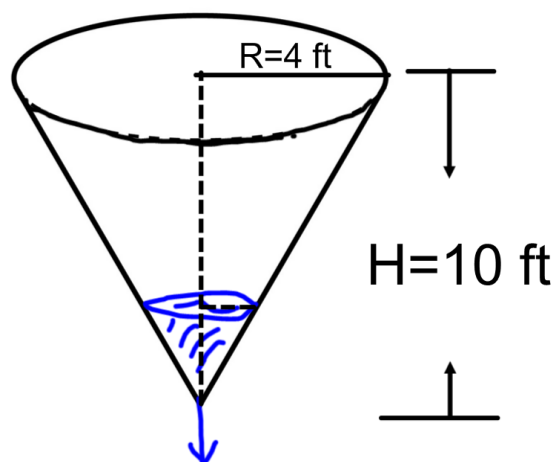


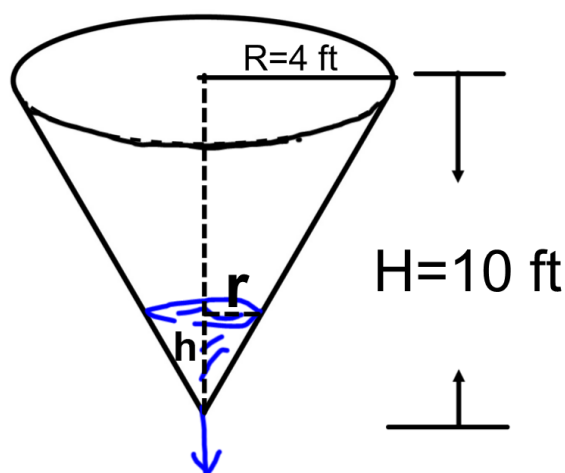
Problem 5

Problem 5

Problem 1 A conical water tank of height 10 feet and radius 4 feet is being drained through a hole in the bottom (See figure below).



- (a) Let r be the radius of the water at depth h . In the figure above, label r and h .



Explanation.

- (b) Find an expression for r in terms of h .

Explanation. $\frac{r}{h} = \frac{R}{H} = \frac{4}{10} = \frac{2}{5} \implies r = \frac{2}{5}h$

- (c) The water in the tank is being drained at a rate of 2 feet³ per minute. What is the rate of change of the water depth when the water depth is 3 feet? [3]

Explanation. Let V be volume of water in the tank. We were given $\frac{dV}{dt} = -2$ feet³/min.

We can relate the volume of water in the tank to the water height by the formula for the volume of a cone: $V = \frac{1}{3}\pi r^2 h$.

We have already found r in terms of h so substituting gives us: $V = \frac{1}{3}\pi \left(\frac{2}{5}h\right)^2 \cdot h = \frac{4}{75}\pi h^3$

$$\begin{aligned} V &= \frac{4}{75}\pi h^3 \\ \frac{dV}{dt} &= \frac{4}{75}\pi \cdot 3h^2 \cdot \frac{dh}{dt} \\ \frac{dV}{dt} &= \frac{4}{25}\pi h^2 \cdot \frac{dh}{dt} \\ -2 &= \frac{4}{25}\pi(3)^2 \cdot \left[\frac{dh}{dt}\right]_{h=3} \\ \left[\frac{dh}{dt}\right]_{h=3} &= -\frac{25}{18\pi} ft/sec \end{aligned}$$